ISA/IEC 62443-3-3 / 4-2 • SL-3 TESTING CHEAT SHEET

Rockwell/Allen-Bradley 1769-AENTR Compact I/O Dual-Port EtherNet/IP Adapter (native Modbus-TCP socket bridge) · Component type: **Embedded Device (EDR) - EtherNet/IP remote-I/O adapter with open-socket Modbus/EtherNet-IP conversion** · Target SL = SL 3

Where to work	Configure in **Studio 5000 Logix Designer** (the adapter + 1769 I/O appear under the CompactLogix I/O tree via the **Add-On Profile** - Module Properties: **Electronic Keying**, connection/RPI); IP is set with the **BOOTP-DHCP Tool** / rotary switches / **FactoryTalk Linx**. **Modbus TCP bridging is native** via the **Socket Interface**: the controller opens TCP sockets through the adapter's **Socket Object** using CIP generic **MSG** instructions (create/connect/write/read/close) to speak Modbus TCP (502) to third-party devices - see Rockwell **ENET-AT002**. The embedded **web server** (HTTP) must be **disabled after commissioning**. Access control is enforced at the **CompactLogix controller + FactoryTalk Security**, not on the adapter.
Ports / paths	**EtherNet/IP:** CIP explicit **TCP 44818**, implicit I/O **UDP 2222**, ListIdentity **UDP 44818**; embedded web **HTTP 80** (**no HTTPS, no CIP Security**); **BOOTP/DHCP UDP 67/68**; **SNMP UDP 161** (if enabled). **Socket-interface / Modbus bridge:** raw TCP via the Socket Object out to third-party devices, typically **Modbus TCP 502** (unauthenticated). Capacity: **96 TCP / 128 CIP** connections (sockets consume from this pool - budget + monitor).
Firmware / CVEs	1769-AENTR = Compact I/O dual-port EtherNet/IP adapter (DLR-capable), rotary-switch addressed; lets CompactLogix reuse 1769 I/O. No CIP Security on this adapter (migration path for a secured conduit: 1756-EN4TR / CIP-Security-capable devices). Patch firmware via **ControlFLASH Plus** and track Rockwell PSA / CISA advisories (the AENTR family has prior web-interface advisories, e.g. ICSA-21-063-01 on the 1734-AENTR - confirm the 1769-AENTR firmware is current and its web server disabled).
How to read	Default read-only (Studio 5000 read-only project review; read-only CIP/EtherNet-IP + Modbus enumeration). Each row = native step-by-step + command-level external-tool validation + pass. Write/exploit tools = isolated env + written authorization. Legacy adapter: many rows are Compensating / N-A-at-device; the SOCKET INTERFACE gets extra scrutiny.

FR 1 — Identification & Authentication Control (IAC)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / platform
SR/CR 1.1	Human user identification & auth	The adapter has **no native user accounts**. Human access is authenticated at the **CompactLogix controller + FactoryTalk Security** (who may go online / download the I/O + socket logic) - confirm that is enforced. The embedded **web server** is an unauthenticated exposure - **disable it** after commissioning and patch firmware. **Test:** browse `http://` -> web disabled/patched; going online to change I/O or socket MSG logic requires FactoryTalk/controller credentials. **Ext. tools:** FactoryTalk Security (policy review); `curl -i http:///` (expect closed); `nmap -p 80,44818` .	Engineering authenticated at controller/FactoryTalk; adapter web disabled + patched; no anonymous config.	EDR - no device auth; compensating.
SR/CR 1.2	Device & software-process I&A;	EtherNet/IP CIP is **unauthenticated** (no CIP Security). Set **Electronic Keying = Exact Match** in Module Properties so the controller binds the exact adapter identity/revision; restrict CIP peers by segmentation. **Socket note:** the socket interface is a controller-initiated raw-TCP client to third-party Modbus devices with **no device/process authentication** - constrain it by destination allow-list + segmentation. **Test:** from an unauthorized host, issue CIP ListIdentity / open a class-3 connection -> blocked by segmentation; confirm socket connections only reach approved Modbus targets. **Ext. tools:** Studio 5000 (Electronic Keying + socket MSG destinations); `nmap --script enip-info -p 44818` ; Wireshark `enip || cip || modbus`.	Exact-match keying binds identity; socket destinations constrained; unauthenticated CIP/Modbus compensated by keying + segmentation.	No CIP Security - Compensating.
SR/CR 1.3	Account management	No accounts on the adapter; manage engineering identities in **FactoryTalk Security** (least-privilege, named users, no shared logins) and keep custody of the Studio 5000 project (which holds the I/O config + socket MSG logic + Modbus destinations). **Ext. tools:** FactoryTalk Security (user/role export); project-custody record.	Named least-privilege engineering accounts (FactoryTalk); project protected.	

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / platform
SR/CR 1.4	Identifier management	Confirm a unique device identity: unique **IP / node address** (rotary switches or BOOTP/DHCP reservation), unique CIP identity (product code/serial); and that each socket-connected third-party device has a unique, documented IP. No reuse across cells. **Ext. tools:** Studio 5000 (module identity + socket destination list); `nmap --script enip-info -p 44818` ; network inventory.	Unique device + socket-peer identifiers; no reuse.	
SR/CR 1.5 / 1.7	Authenticator management & strength	The adapter stores **no device credentials**; authenticator strength is enforced at **FactoryTalk Security** (engineering). Modbus TCP over the socket interface is itself credential-less - document that and compensate with segmentation. **Ext. tools:** FactoryTalk Security (password policy); note Modbus has no authenticator.	Strong FactoryTalk credentials; Modbus-credential gap documented + compensated.	Adapter/Modbus: no credentials.
SR/CR 1.6	Wireless access management	**N/A.** Wired Compact I/O adapter (in-cabinet) with no native wireless interface. **Ext. tools:** Confirm no wireless; upstream network governs.	N/A - justification recorded (wired device).	N/A
SR/CR 1.8 / 1.9	PKI certificates & key strength	**N/A / limited at device.** 1769-AENTR has no device PKI/TLS (HTTP-only, no HTTPS, no CIP Security); the socket interface is raw TCP with no TLS. Record N/A; provide confidentiality/integrity via segmentation, or migrate a secured conduit to a CIP-Security-capable path. **Ext. tools:** Confirm no configurable device certificate; Wireshark (web = HTTP cleartext; CIP + Modbus unauthenticated).	N/A/limited device PKI; compensating segmentation upstream.	N/A/Partial.
SR/CR 1.10	Authenticator feedback	**N/A at adapter** (no device login). Confirm FactoryTalk logon masks the secret (no cleartext echo). **Ext. tools:** Visual verification (FactoryTalk).	Authenticators obscured at FactoryTalk (N/A at adapter).	N/A at device.
SR/CR 1.11	Unsuccessful login attempts	The adapter has **no login / no lockout**. Enforce lockout at **FactoryTalk Security** (engineering); remove the adapter's exposure by disabling the web server + segmentation; monitor failed CIP/Modbus/socket attempts. **Ext. tools:** FactoryTalk Security (lockout policy); OT IDS on the cell segment.	Lockout at FactoryTalk; adapter web disabled; attempts monitored.	Adapter: no lockout - Compensating.
SR/CR 1.12	System use notification	No banner on the adapter/embedded web; enforce the approved use notice on the Studio 5000 EWS (GPO) and the SCADA HMI (compensating). **Ext. tools:** EWS `gpresult /h gp.html`; HMI banner review.	Use notice at EWS/HMI (compensating at adapter).	Compensating.
SR/CR 1.13	Access via untrusted networks	Segment the adapter into a cell zone; firewall **44818/2222/80/161**; no EtherNet/IP from untrusted networks; VPN for cross-zone engineering. The **socket/Modbus 502** path to third-party devices crosses toward another network - firewall it separately and keep those Modbus targets in a controlled segment. **Test:** from an untrusted VLAN, reach 44818 / the socket-Modbus 502 targets -> must fail. **Ext. tools:** `nmap -Pn -sTU -p 44818,2222,80,161,502` from the untrusted side (expect filtered); firewall rule review.	Adapter + socket-Modbus path unreachable from untrusted networks; brokered via VPN + firewall.	

FR 2 — Use Control (UC)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / platform
SR/CR 2.1	Authorization enforcement	The adapter performs **no independent authorization** - it presents I/O and relays socket traffic under the owning controller. Confirm authorization is enforced at the **CompactLogix controller + FactoryTalk Security** (roles: go-online / download / edit), that **Electronic Keying** prevents a mismatched module, and that the socket MSG logic only writes approved Modbus registers. **Test:** as a read-only FactoryTalk user, attempt to download / change the I/O or socket logic -> denied. **Ext. tools:** FactoryTalk Security (role review); Studio 5000 (keying + socket MSG write scope).	Least privilege enforced at controller/FactoryTalk; keying blocks mismatched modules; socket writes scoped.	Delegated to controller.
SR/CR 2.2	Wireless Use Control	**N/A.** No wireless interface on this device; wireless access is managed at the network layer.	N/A - justification recorded (not assessed as a control).	N/A
SR/CR 2.3	Use control for portable/mobile devices	Control the Studio 5000 laptop and removable media (OS device-control); only authorized projects/firmware loaded; the 1769-AENTR sits in a cabinet - control physical access. **Ext. tools:** EWS removable-media GPO; physical inspection (cabinet).	Engineering media controlled; adapter physical access controlled.	

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / platform
SR/CR 2.4 (+ EDR 2.4)	Mobile code	No general-purpose mobile code; confirm only Rockwell firmware runs and the embedded web content is vendor read-only - **disable the web server**. The socket interface relays raw bytes but executes no downloaded code. **Ext. tools:** Studio 5000 (firmware review); confirm web server disabled.	Only trusted vendor firmware runs; web disabled; no untrusted mobile code.	EDR-specific.
SR/CR 2.5	Session lock	**N/A at adapter** (no interactive session). Enforce workstation lock on the Studio 5000 EWS (OS screen lock/GPO). **Ext. tools:** EWS inactivity-lock GPO.	EWS locks on inactivity (N/A at adapter).	Compensating.
SR/CR 2.6 / 2.7	Remote session termination & concurrency	Confirm CIP + socket connections can be closed and the adapter **bounds concurrency** - **96 TCP / 128 CIP** connections - so it cannot be exhausted; the controller closes stale sockets (socket-close MSG) and bounds concurrent Modbus sessions. **Ext. tools:** Studio 5000 (socket create/close logic; connection count); `nmap --script enip-info`; Wireshark (teardown).	CIP + socket connections bounded/terminable; stale sockets closed.	
SR/CR 2.8 / 2.12	Auditable events & non-repudiation	The adapter has **no native security/syslog log**. Capture configuration-change attribution at the **CompactLogix controller + FactoryTalk AssetCentre**, and log CIP/Modbus/socket connection + write events with a **CIP/Modbus-aware OT IDS**; forward to the SIEM. **Ext. tools:** FactoryTalk AssetCentre (change/audit); OT IDS (CIP/Modbus) on a SPAN + SIEM.	Changes attributable at controller/AssetCentre; adapter/socket events captured by IDS.	No native syslog - Compensating.
SR/CR 2.9 / 2.10	Audit storage capacity & failure handling	Because logging is off-device (AssetCentre / IDS / SIEM), confirm those are sized/retained and alert on collection failure so adapter/socket events are not silently lost. **Ext. tools:** AssetCentre / SIEM retention + source-silent alert review.	Off-device logs sized/retained; audit-failure alerting defined.	Compensating.
SR/CR 2.11	Timestamps	The adapter has no user RTC; event times come from the **controller / SCADA**. Confirm the controller and IDS/SIEM are time-synced (NTP/CIP time) so events correlate. **Ext. tools:** Studio 5000 / controller time (CIP); `nmap -sU -p 123` / Wireshark `ntp`.	Reliable synchronized time at controller/IDS; events correlate.	
EDR 2.13	Physical diagnostic & test interfaces	Control physical interfaces: adapter in a **locked cabinet**; protect the **rotary address switches** (set + record node) and the two Ethernet ports; disable the web server and any unused service (SNMP) so no diagnostic backdoor remains. **Ext. tools:** Physical inspection (cabinet, rotary-switch cover); `nmap -sU -p 161` (SNMP off if unused).	Physical/diagnostic interfaces controlled; web + unused services disabled.	EDR-specific.

FR 3 — System Integrity (SI)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / platform
SR/CR 3.1 / 4.1	Communication integrity & confidentiality	EtherNet/IP CIP, the socket-interface **Modbus TCP**, and the embedded web are **unauthenticated cleartext** (no CIP Security/HTTPS/TLS) - integrity/confidentiality come from **containment**. Keep CIP/I/O inside a segmented cell zone and confine the socket-Modbus targets to a controlled segment behind a firewall; no uncontrolled cleartext control crosses a zone boundary. **Test:** confirm CIP (44818/2222) and socket-Modbus (502) stay within controlled segments (no cleartext crossing the boundary uncontrolled). **Ext. tools:** Wireshark `enip || cip || modbus` (cleartext, in-zone only); firewall rule review.	Control + socket-Modbus comms contained/segmented; no uncontrolled cleartext across boundaries.	No CIP Security - Compensating.
EDR 3.2	Protection from malicious code (component)	Confirm only Rockwell firmware runs on the adapter. Patch via **ControlFLASH Plus** (newer Rockwell images are digitally signed and verified on load); obtain firmware from Rockwell PCDC; track Rockwell PSA / CISA advisories. **Ext. tools:** ControlFLASH Plus (version + signature); Studio 5000 (module firmware); Rockwell PSA / CISA review.	Only vendor firmware runs; patched; provenance/signature verified.	EDR-specific.
SR/CR 3.3	Security functionality verification	After any change re-verify: web server disabled, unused services off (SNMP), Electronic Keying set, firmware patched, segmentation intact, socket-Modbus destinations still allow-listed, against the hardening baseline; compare the Studio 5000 project. **Ext. tools:** Studio 5000 (settings + socket destinations vs baseline); `nmap -sTU -p 44818,2222,80,161,502` (only required reachable).	Security settings match the hardening baseline after change.	

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / platform
SR/CR 3.4	Software & information integrity	The adapter/I/O config + socket MSG logic + Modbus map live in the **Studio 5000 project** - protect it (host ACLs, version control); Electronic Keying + controller download-control prevent unauthorized change; hash the project export to detect drift. **Ext. tools:** Studio 5000 (online-vs-offline compare); `certutil -hashfile .ACD SHA256` for drift.	Config + socket-logic integrity protected; unauthorized change prevented/detected.	
SR/CR 3.5	Input validation	Confirm the adapter tolerates malformed **CIP / EtherNet/IP** frames and the socket path tolerates malformed **Modbus** responses without faulting or dropping I/O. **ISOLATED only:** fuzz on a bench unit and watch connection/I/O/socket health. **Ext. tools:** CIP/EtherNet-IP + Modbus fuzzer (`boofuzz`) on a bench unit; Wireshark; watch adapter status LEDs + connection/socket state.	No fault or loss of I/O on malformed input.	Do NOT fuzz an in-service unit.
SR/CR 3.6	Deterministic output	Verify configured **fault/fallback behavior**: 1769 output modules assume their configured safe state (fault action / hold-last) on CIP connection loss/timeout; and the controller handles socket-Modbus timeouts deterministically (no stuck outputs from stale Modbus data). **Ext. tools:** Studio 5000 (output fault action / connection timeout; socket-MSG timeout handling); simulate comm loss -> defined output state.	Outputs go to a defined/fail-safe state on connection or Modbus-socket loss.	
SR/CR 3.7	Error handling	Confirm graceful fault handling: adapter **module/network status LEDs** + controller connection/socket diagnostics (MSG error codes) report faults without unsafe behavior or info leak. **Ext. tools:** Studio 5000 (module + socket-MSG diagnostics); adapter LED/status review.	Graceful fault handling; no unsafe state or info leak.	
SR/CR 3.8	Session integrity	**N/A / limited.** No authenticated/encrypted session on the adapter or the socket-Modbus path (no CIP Security/TLS); integrity is provided by containment. Record N/A + compensating segmentation. **Ext. tools:** Wireshark (confirm in-zone containment; no replay path across boundary).	N/A/limited at device; compensating segmentation recorded.	N/A/Partial.
SR/CR 3.9	Protection of audit information	Since audit is off-device, protect the AssetCentre / IDS / SIEM store and restrict who can read/clear it. **Ext. tools:** AssetCentre / SIEM access review.	Audit records protected from unauthorized read/modify/delete.	
EDR 3.10	Support for updates	Update adapter firmware via **ControlFLASH Plus** with firmware from Rockwell; verify signature/version; plan in a maintenance window with rollback to the prior revision; apply current advisories. **Ext. tools:** ControlFLASH Plus (update flow + signature); Rockwell PCDC provenance.	Controlled signed updates from a verified source, patched, with rollback.	EDR-specific.
EDR 3.11	Physical tamper resistance/detection	Keep the adapter in a locked/sealed cabinet; control the rotary switches, Ethernet ports, and the 1769 bus connection; config changes are visible in Studio 5000 / AssetCentre. **Ext. tools:** Physical inspection (cabinet/seal, rotary cover); AssetCentre change review.	Cabinet secured; interfaces controlled; changes logged.	EDR-specific.
EDR 3.12	Provisioning product-supplier roots of trust	Newer Rockwell firmware is **digitally signed** and verified by ControlFLASH Plus (product-supplier root of trust); confirm signed firmware / provenance for the loaded revision. Older revisions may predate signed images - record and compensate with provenance discipline (3.10), patching, and segmentation. **Ext. tools:** ControlFLASH Plus (signature verify); Rockwell PCDC provenance.	Signed-firmware supplier RoT where supported; provenance verified; gaps recorded.	EDR-specific/Partial.
EDR 3.13	Provisioning asset-owner roots of trust	**Partial/N-A at device.** The 1769-AENTR provides no asset-owner certificate provisioning (no CIP Security/PKI); the socket path is raw TCP. Record the gap; asset-owner trust is enforced at the network layer (VPN/firewall/segmentation); migrate a secured conduit to a CIP-Security-capable device where required. **Ext. tools:** Confirm no device certificate/CIP-Security provisioning; segmentation/migration review.	No asset-owner RoT at device; compensating controls / migration path recorded.	Partial/N-A.
EDR 3.14	Integrity of the boot process	Signed-firmware load via ControlFLASH Plus provides load-time integrity on supported revisions; there is **no published secure-boot attestation** for the adapter. Record the gap; mitigate via controlled firmware provenance (3.10), patching, physical control, and monitoring. **Ext. tools:** ControlFLASH Plus (firmware signature/version); confirm no secure-boot attestation.	Load-time firmware integrity where supported; no attestation - compensating controls recorded.	EDR-specific/Partial.

FR 4 — Data Confidentiality (DC)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / platform
SR/CR 4.2	Information persistence	On decommission/repurpose, **reset the adapter to out-of-box** (clear IP/config) and clear the I/O config + socket MSG logic + Modbus destinations from the Studio 5000 project so network config and target lists are not recoverable; sanitize any removable media. **Ext. tools:** Adapter reset-to-defaults; Studio 5000 (remove module + socket logic).	Adapter config + socket/Modbus destinations purged on decommission or reuse.	
SR/CR 4.3	Use of cryptography	**N/A / limited.** No comms crypto on the adapter or socket path (no HTTPS/CIP Security/TLS). Rely on segmentation and TLS at the SCADA/EWS layer; **disable the embedded web**. Where confidentiality is required on the conduit, migrate to a CIP-Security-capable path. **Ext. tools:** Confirm no device crypto; confirm web disabled; Wireshark (no cleartext across boundary).	No device crypto - compensating TLS/segmentation upstream; web disabled.	N/A/Partial.

FR 5 — Restricted Data Flow (RDF)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / platform
SR/CR 5.1 / 5.2	Network segmentation & zone boundary	Primary control. Confirm the adapter sits in a segmented cell zone; only required flows cross (CIP 44818/2222 to the owning CompactLogix). The **socket interface reaches out to Modbus TCP 502 devices** - treat that as a conduit: firewall it, keep the Modbus targets in a controlled segment, and allow only the intended socket flows. **Ext. tools:** `nmap -sTU -p 44818,2222,80,161,502` per zone; firewall/conduit rule review.	Adapter in a segmented zone; socket-Modbus conduit firewalled to a controlled target segment; only required flows cross.	
SR/CR 5.3	General-purpose person-to-person comms restrictions	**N/A.** Single-purpose I/O adapter / socket bridge with no general-purpose person-to-person messaging. **Ext. tools:** Confirm no such services.	N/A - justification recorded.	N/A
SR/CR 5.4	Application partitioning	The socket interface effectively bridges the EtherNet/IP cell to a Modbus network - confirm those are **partitioned** (separate segments + firewall) and only the mapped socket flows cross; the adapter itself is single-purpose. **Ext. tools:** Segmentation review of the CIP cell vs the socket-Modbus target segment; `nmap` per segment.	CIP cell and socket-Modbus network partitioned; only mapped flows cross.	

FR 6 — Timely Response to Events (TRE)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / platform
SR/CR 6.1 / 6.2	Audit log accessibility & continuous monitoring	The adapter has no native syslog, so monitoring is off-device: confirm a passive **CIP/Modbus-aware OT IDS** (Nozomi/Dragos) on the cell segment + the socket-Modbus conduit captures connection/write/CVE-exploitation events and forwards to the SIEM; **FactoryTalk AssetCentre** holds config-change records that are exportable. **Ext. tools:** Passive OT IDS (CIP/Modbus-aware) on a SPAN; SIEM; AssetCentre change export.	Adapter/socket events monitored off-device in near-real-time; changes exportable.	No native syslog - Compensating.

FR 7 — Resource Availability (RA)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / platform
SR/CR 7.1	Denial-of-service protection	Confirm DoS resilience: the adapter is sensitive to EtherNet/IP floods and to **socket/connection exhaustion** (96 TCP / 128 CIP pool - the socket bridge draws from it) - use cell-firewall rate controls and **DLR** for media resiliency; ensure the controller closes stale sockets so the pool is not exhausted. **Ext. tools:** ISOLATED bench: `hping3 --flood -p 44818` / CIP + socket connection storm; watch connection/socket pool + DLR + I/O state.	Resources bounded (connection pool protected); I/O continues under load; ring resiliency confirmed.	

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / platform
SR/CR 7.2	Resource management	Confirm bounded resources: budget the **96 TCP / 128 CIP** connection pool across I/O + sockets, size RPI appropriately, and cap concurrent socket-Modbus sessions so the bridge cannot starve I/O. **Ext. tools:** Studio 5000 (connection/RPI + socket count budget); `nmap --script enip-info`.	Connection pool budgeted; no exhaustion-based loss of I/O.	
SR/CR 7.3 / 7.4	Backup & recovery	Confirm the **Studio 5000 project** (holds adapter/I/O config + socket logic + Modbus map) is backed up (versioned, offline) and a restore is tested on a spare adapter for a known-good rebuild. **Ext. tools:** Backup repository; `certutil -hashfile .ACD SHA256`; restore test (spare unit).	Backups current and recovery validated.	
SR/CR 7.5	Emergency Power	**N/A.** Emergency/backup power is provided at the facility/UPS level, not the device; confirm the device retains its security state across a power transfer.	N/A - justification recorded (not assessed as a control).	N/A
SR/CR 7.6 / 7.7	Network & security settings + least functionality	Harden: **disable the embedded web server**, disable unused services (SNMP), set Electronic Keying, patch firmware; only enable the socket interface if required and cap the number of sockets; restrict the socket-Modbus destinations to the required peers. **Ext. tools:** Adapter web (services off); Studio 5000 (keying + socket-enable/count + destinations); `nmap -sTU -p 80,161,44818,2222,502` (only required reachable).	Least functionality; web + unused services disabled; keying set; socket use minimized; patched.	
SR/CR 7.8	Component inventory	Inventory the adapter (1769-AENTR catalog + series + firmware) and every socket-connected Modbus target so vulnerabilities track against **Rockwell PSA / CISA** advisories. **Ext. tools:** `nmap --script enip-info -p 44818` ; Studio 5000 inventory (+ socket destination list); Rockwell PSA feed.	Accurate adapter/firmware + socket-peer inventory maintained; advisories tracked.	

External Test Tools — how to use

Tool / method	Verifies	How to use — steps / example	Constraints
Studio 5000 Logix Designer (AOP)	Native config, keying & socket-logic review (all FRs)	Open the project read-only; the adapter + 1769 I/O appear under the CompactLogix I/O tree (Add-On Profile) - review Module Properties (Electronic Keying, connection/RPI), the socket-interface MSG logic (create/connect/write/read/close + destinations), module diagnostics, firmware. Export as the baseline.	Read/authorized project copy.
EtherNet/IP Socket Interface App Technique (ENET-AT002)	Understand + audit the Modbus socket bridge (1.2, 3.1, 5.x)	Reference for the Socket Object (create_socket/open_connection/send/receive/delete via CIP generic MSG); use it to confirm the socket-Modbus implementation, destinations, and close/timeout handling.	Reference.
FactoryTalk Security / AssetCentre	Engineering I&A, authorization & audit (1.1, 1.3, 2.1, 2.8, 6.x)	Where human access is actually enforced (roles: go-online/download); AssetCentre holds config-change/audit records - export them.	Read-only review.
BOOTP-DHCP Tool / FactoryTalk Linx	IP assignment & browse (1.2, 1.4)	Set/reserve the adapter IP; browse the EtherNet/IP tree; confirm device identity. Prefer a DHCP reservation or disable BOOTP after commissioning.	Authorized; commissioning.
ControlFLASH Plus	Firmware update, signature & provenance (3.2, 3.10, 3.12)	Update the adapter to current firmware from Rockwell (PCDC); confirm signature/version; keep the prior revision for rollback.	Maintenance window.
Embedded web server / curl	Web-exposure check (1.1, 2.4, 7.6)	`curl -i http:///`; confirm the web server is DISABLED after commissioning; no unauthenticated config.	Read-only; authorized.
nmap	Port/service exposure, EtherNet/IP ID & segmentation (1.x, 3.1, 5.x, 7.x)	`nmap -sTU -p 44818,2222,80,161,502` ; `nmap --script enip-info -p 44818` ; scan per zone incl. the socket-Modbus targets.	Read-only; authorized.
Wireshark / tcpdump	Comms integrity, identity & socket bridge (1.2, 3.1, 3.8)	Filter `enip || cip || modbus`; confirm cleartext CIP + socket-Modbus stay in controlled segments; inspect the socket connection setup/teardown.	Passive; SPAN/tap only.
Modbus client / CIP-Modbus fuzzer (isolated)	Socket-bridge validation & input handling (1.2, 3.5)	Read-only Modbus queries to confirm which targets answer; `boofuzz` malformed Modbus responses to a bench unit; watch socket/I/O health.	ISOLATED bench; written authorization.

Tool / method	Verifies	How to use — steps / example	Constraints
Passive OT IDS (CIP/Modbus-aware)	Monitoring (2.8, 6.x) - replaces native syslog	Nozomi/Dragos on a SPAN to capture CIP/Modbus/socket connection + write + CVE-exploitation attempts; forward to the SIEM.	Passive; coordinate with SOC.
Rockwell PSA / CISA advisories	Vuln tracking (3.2, 3.10, 7.8)	Track 1769-AENTR / AENTR-family advisories (e.g. ICSA-21-063-01 on the 1734-AENTR) + required firmware.	Reference.

Controlled - Project Use · Independent Assurance Authority · ISA/IEC 62443-3-3 / 4-2 · Product-specific companion to the 62443 assessment tool; confirm exact menu paths / CLI against the deployed firmware and the project hardening baseline. ISOLATED tools run only in a representative/lab environment with written authorization.